

Survey Paper Workflow for Object Counting Literature Review

Combined notes from the conversation on Google Scholar, Codex, Claude Code, MCP tools, and seed-paper collection

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Core recommendation: Use Google Scholar manually for seed discovery and verification only. Use Claude Code or Codex with legitimate scholarly APIs such as OpenAlex, Semantic Scholar, Crossref, and arXiv to automate bulk metadata collection, deduplication, literature-matrix creation, grouping, and outline generation.

Document overview

1. Can Codex access Google Scholar?
2. Can Claude Code do it?
3. What are Google Scholar MCP servers and gs-skills?
4. How to handle 10,000 Google Scholar results.
5. Whether manual Google Scholar search is still needed.
6. Best way to collect seed papers for an object counting survey across modalities.
7. A ready-to-use Claude Code/Codex prompt.
8. Appendix: source links mentioned and literature-matrix fields.

1. Can Codex access Google Scholar?

Question: Can I use Codex to access Google Scholar and retrieve information to save time?

Answer: Codex can help automate the literature-review workflow, but it should not be used to scrape Google Scholar directly. Google Scholar is restrictive about automated access, automated querying can trigger CAPTCHA or blocking, and Scholar does not provide an official bulk-access API for all search records.

A safer workflow is:

9. Use Google Scholar manually for seed papers.
10. Export citations to BibTeX/RIS, or save DOI/title/PDF links.
11. Use Codex to automate metadata collection from allowed sources such as OpenAlex, Semantic Scholar, Crossref, arXiv, PubMed, or exported citations from IEEE/ACM if relevant.
12. Use Codex to create literature matrices, .bib files, theme groupings, duplicate removal, LaTeX tables, and survey outlines.

Safe pipeline:

```
Google Scholar/manual search -> collect seed papers -> export BibTeX
    -> Codex script using OpenAlex/Semantic Scholar/Crossref/arXiv
    -> CSV/Excel literature matrix + BibTeX file
    -> Thematic grouping + survey outline
```

Example Codex prompt:

```
Write a Python script that takes a list of seed paper titles from a CSV file,
queries OpenAlex and Semantic Scholar for metadata, retrieves abstracts,
publication year, venue, DOI, citation count if available, and influential
references, removes duplicates, and saves the output as literature_matrix.xlsx
and references.bib.
```

2. Can Claude Code do this?

Question: What about using Claude Code?

Answer: Claude Code has the same practical and policy issue: it is useful for building scripts and organizing the survey workflow, but it should not be used to bypass Google Scholar restrictions or scrape Scholar at scale.

Recommended Claude Code workflow:

1. Manually search Google Scholar for seed papers.
2. Export citations from Scholar as BibTeX/RIS.
3. Put the seed titles/DOIs into a CSV file.
4. Ask Claude Code to write scripts that query OpenAlex, Semantic Scholar, Crossref, arXiv, etc.
5. Generate literature_matrix.xlsx, references.bib, paper_summary.md,

and survey_outline.tex.

Example Claude Code prompt:

```
Create a Python pipeline for a survey paper.
Input: seed_papers.csv with title, authors, DOI if available.
Use OpenAlex, Semantic Scholar, Crossref, and arXiv APIs to retrieve metadata,
abstracts, citation counts, references, and related papers.
Deduplicate by DOI and fuzzy title matching.
Save outputs as literature_matrix.xlsx, references.bib, and survey_outline.md.
```

3. What about Google Scholar MCP servers or gs-skills?

Question: What is meant by a “Direct Method: Connect an MCP Server” for Claude Code to query Google Scholar?

Answer: Those are real third-party tools or community integrations, but they do not make bulk Google Scholar retrieval officially approved or risk-free. MCP lets Claude Code connect to external tools, APIs, databases, and browser-automation workflows. Some Google Scholar MCP tools wrap scraping libraries or automate a browser to access Scholar pages.

Important distinction:

Question	Practical answer
Can Claude Code technically query Scholar through an MCP tool?	Yes, some community tools can automate Scholar-like searches or browser navigation.
Does this mean Google Scholar officially permits bulk scraping?	No. The integration being installable does not make the workflow officially approved.
Is it reliable for a full survey paper?	Not ideal. It may trigger CAPTCHA, blocking, rate limits, or incomplete results.
Best use of such tools	At most, light browsing assistance or small-scale checking; not the main bulk collection pipeline.

Practical recommendation: Treat Google Scholar MCP tools as risky for large-scale survey-paper collection. Use them, at most, as a light helper for manual browsing. For the real pipeline, use scholarly APIs designed for automated access.

4. How to handle 10,000 Google Scholar results

Problem: Searching “object counting” returns thousands of results, and it is impossible to inspect all of them manually.

Solution: create a screening pipeline instead of reading everything.

```
Broad topic: Object Counting
-> Narrow subtopics
-> Search 5-10 targeted queries
-> Collect top 50-100 papers per query
-> Deduplicate
-> Screen titles/abstracts
-> Read 40-100 papers deeply
```

Better targeted queries than just “object counting”:

- "object counting" "computer vision" survey
- "object counting" "few-shot"
- "object counting" "zero-shot"
- "open-vocabulary object counting"
- "class-agnostic object counting"
- "exemplar-based object counting"

- "dense object counting"
- "visual counting" "foundation models"
- "video object counting"
- "object counting" "SAM"
- "object counting" "vision-language models"

Useful thematic divisions:

Theme	What to collect
Traditional object counting	Early density-map, regression-based, and detection-based counting methods.
Few-shot / exemplar-based counting	Methods that use example boxes, patches, or exemplars.
Class-agnostic counting	Methods that count unseen object categories.
Zero-shot counting	Methods that count from text/class names without examples.
Open-vocabulary counting	Vision-language and foundation-model-based counting.
Video object counting	Temporal counting, tracking, identity consistency, and re-identification.
Datasets and metrics	FSC-147, FSCD-147, CountBench, OmniCount, CARPK, ShanghaiTech, TAO-Count, VideoCount, etc.
Applications	Agriculture, traffic analysis, wildlife monitoring, healthcare, microscopy, manufacturing, and remote sensing.

Screening rule:

Keep	Remove
Directly about object counting, visual counting, or counting benchmarks.	Only mentions counting as a minor application.
Introduces a method, dataset, benchmark, metric, or survey relevant to counting.	Generic object detection papers without counting evaluation.
Strong venue, high citations, recent work, or directly related to open-vocabulary/foundation-model counting.	Unrelated to image/video/visual modalities.
Useful domain paper showing counting applications.	Too domain-specific unless it supports an application section.

5. Do you still need manual Google Scholar search?

Question: Do I still need Google Scholar manual search?

Answer: Yes, but only for seed discovery and sanity checking. You do not need to manually search all papers. Scholar is useful for finding the first strong papers, checking “Cited by” and “Related articles,” and verifying recent or highly cited papers. The main collection and organization should be handled through scripts and APIs.

Recommended workflow:

1. Use Google Scholar manually for 30-60 minutes.
-> Find 10-20 strong seed papers.
2. Export/save those seed papers.
-> BibTeX, DOI, title, or PDF links.
3. Use Claude Code/Codex with APIs.
-> OpenAlex, Semantic Scholar, Crossref, arXiv.
4. Expand automatically.
-> Find related papers, citations, references, and newer works.
5. Deduplicate and screen.
-> Reduce 500-1000 collected papers down to 80-150 relevant papers.
6. Read deeply.

-> Focus on the most important 40-80 papers.

Simple summary: Google Scholar = seed and sanity check. Claude Code/Codex + APIs = bulk collection and organization. You = final screening, reading, and writing.

6. Best way to collect seed papers for an object-counting survey across modalities

Survey goal: Understand object counting across modalities, especially images, videos, and possible emerging modalities such as 3D/point-cloud data, remote sensing, microscopy, thermal imagery, depth/RGB-D, and event-camera settings.

Best strategy: Use a modality-first search plan rather than one broad search for “object counting.”

Bucket	Seed paper target
Survey/taxonomy papers	5-10
Image object counting	20-30
Video object counting	10-15
3D / point cloud / remote sensing / microscopy / other modalities	10-15
Dataset and benchmark papers	10-15
Total	55-85 seed papers

6.1 Survey and taxonomy papers

Start here because survey papers expose the taxonomy, major datasets, and foundational references.

- "class-agnostic counting" survey
- "object counting" "computer vision" survey
- "visual counting" survey
- "few-shot object counting" survey
- "open-vocabulary object counting" survey

Collect about 5-10 seed papers from this bucket.

6.2 Image object counting

This will likely be the largest bucket because most object-counting methods begin with images.

- "few-shot object counting"
- "exemplar-based object counting"
- "class-agnostic object counting"
- "zero-shot object counting"
- "open-vocabulary object counting"
- "object counting and detection" "FSCD-147"
- "object counting" "FSC-147"
- "visual counting" "language vision models"
- "object counting" "foundation models"
- "object counting" "SAM"

Collect about 20-30 image-counting seed papers. Include few-shot, exemplar-based, class-agnostic, zero-shot, open-vocabulary, density-based, detection-based, and foundation-model methods.

6.3 Video object counting

Video counting is different from image counting because the goal often involves counting unique objects over time while avoiding duplicate counts when objects disappear and reappear.

- "video object counting"
- "open-world object counting in videos"
- "object counting in videos"
- "video counting" "tracking"
- "video counting" "re-identification"
- "unique instance counting" video
- "TAO-Count" object counting
- "VideoCount" object counting
- "MOT20-Count" object counting
- "CountVid" object counting

Collect about 10-15 video-counting seed papers. Include tracking and re-identification background if they directly support counting over time.

6.4 Other modalities and specialized domains

Use this bucket to answer whether object counting exists beyond ordinary images and videos. These can be extended modalities or specialized domains.

- "3D object counting"
- "object counting" "point cloud"
- "RGB-D object counting"
- "object counting" "depth image"
- "remote sensing" "object counting"
- "aerial image" "object counting"
- "UAV" "object counting"
- "microscopy" "object counting"
- "cell counting" "computer vision"
- "X-ray" "object counting"
- "thermal image" "object counting"
- "event camera" "object counting"

Collect about 10-15 papers. Treat these as extended modality/application sections unless the survey becomes fully multi-modal.

6.5 Dataset and benchmark papers

Search datasets separately because dataset papers often define the evaluation landscape.

- "FSC-147" object counting
- "FSCD-147" object counting
- "CountBench" object counting
- "OmniCount" object counting
- "CARPK" counting
- "ShanghaiTech" counting
- "TAO-Count"
- "VideoCount" object counting

- "MOT20-Count"
- "Science-Count"

For each dataset, try to collect:

- The dataset paper.
- The main benchmark/evaluation paper.
- Two or three methods that report results on that dataset.

Modality	Dataset example	Input prompt	Output	Common metrics
Image	FSC-147	Exemplars / text	Count	MAE, RMSE
Image	FSCD-147	Exemplars / text	Boxes + count	MAE, RMSE, AP
Image	CountBench	Text	Count	Count accuracy
Video	TAO-Count / VideoCount	Text or image prompt	Unique object count	MAE, identity consistency
Other	Remote sensing / microscopy / 3D	Image, volume, point cloud, or depth	Count	MAE, RMSE

6.6 Practical Google Scholar screening rule

For each search query, do not continue through endless pages. Use this stopping rule:

For each query:

- Inspect the top 20 results sorted by relevance.
- Inspect the top 10 results sorted by date.
- Open "Cited by" for 3-5 important papers.
- Open "Related articles" for 3-5 important papers.
- Then stop and move to the next query.

6.7 Recommended five-day seed collection plan

Day	Task	Target
Day 1	Search survey/taxonomy papers.	Collect 5-10 papers.
Day 2	Search image object counting: few-shot, class-agnostic, zero-shot, open-vocabulary, foundation-model counting.	Collect 20-30 papers.
Day 3	Search video object counting: video counting, open-world video counting, CountVid, VideoCount, TAO-Count.	Collect 10-15 papers.
Day 4	Search other modalities: 3D, point cloud, remote sensing, microscopy, cell counting, X-ray/science videos.	Collect 10-15 papers.
Day 5	Use Claude Code/Codex to expand, deduplicate, and organize everything.	Produce matrix, BibTeX, and outline.

6.8 Possible survey titles

- A Survey of Object Counting in Computer Vision: From Exemplar-Based Methods to Open-Vocabulary and Video Counting
- Object Counting Across Modalities: A Survey of Image, Video, Open-Vocabulary, and Emerging 3D Counting Methods
- Object Counting in the Era of Foundation Models: A Survey Across Images, Videos, and Open-Vocabulary Modalities

7. Ready-to-use Claude Code/Codex prompt

You can give this prompt to Claude Code or Codex after collecting your seed papers:

I am preparing a survey paper on object counting across modalities, with emphasis on image object counting, video object counting, zero-shot counting, few-shot/exemplar-based counting, class-agnostic counting, open-vocabulary counting, and emerging modalities such as 3D/point-cloud, remote sensing, microscopy, thermal, RGB-D/depth, and event-camera counting.

Create a literature review pipeline that:

1. Takes seed_papers.csv with columns: title, authors, year, DOI, venue, URL, modality.
2. Uses OpenAlex, Semantic Scholar, Crossref, and arXiv APIs to retrieve metadata, abstracts, citation counts where available, references, related papers, and BibTeX.
3. Deduplicates papers by DOI and fuzzy title matching.
4. Groups papers into themes:
 - survey/taxonomy
 - traditional image counting
 - density-map/regression-based counting
 - detection-based counting
 - few-shot/exemplar-based counting
 - class-agnostic counting
 - zero-shot counting
 - open-vocabulary/foundation-model counting
 - video object counting
 - 3D/point-cloud/depth counting
 - remote sensing/aerial/UAV counting
 - microscopy/cell/scientific counting
 - datasets and benchmarks
 - applications
5. Saves outputs as:
 - literature_matrix.xlsx
 - references.bib
 - grouped_papers.md
 - survey_outline.md
 - missing_metadata_report.csv
6. Adds a relevance score and a reason for keeping or excluding each paper.
7. Produces a draft survey outline with sections, subsections, and suggested papers to cite in each subsection.

8. Literature matrix fields to use

- Paper title
- Authors
- Year
- Venue
- DOI
- URL
- Modality: image / video / 3D / remote sensing / microscopy / other
- Task type: density-based / detection-based / few-shot / zero-shot / open-vocabulary / tracking-based
- Input: image / video / text prompt / exemplars / point cloud / depth / event stream
- Output: count only / boxes + count / masks + count / tracks + count
- Dataset(s)
- Metrics
- Key contribution
- Main limitation
- Relevance: high / medium / low
- BibTeX copied? yes/no
- PDF saved? yes/no
- Notes for survey section

9. Source links mentioned during the conversation

These are the source links that were referenced in the conversation. They are included for follow-up verification, not as a replacement for your own literature screening.

- [Google Scholar Help](https://scholar.google.com/intl/en/scholar/help.html) - <https://scholar.google.com/intl/en/scholar/help.html>
- [Google Scholar robots.txt](https://scholar.google.com/robots.txt) - <https://scholar.google.com/robots.txt>
- [Claude Code MCP docs](https://code.claude.com/docs/en/mcp) - <https://code.claude.com/docs/en/mcp>
- [Claude Code overview](https://code.claude.com/docs/en/overview) - <https://code.claude.com/docs/en/overview>
- [OpenAlex API documentation](https://developers.openalex.org/api-reference/introduction) - <https://developers.openalex.org/api-reference/introduction>
- [OpenAlex website](https://openalex.org/) - <https://openalex.org/>
- [Semantic Scholar API](https://www.semanticscholar.org/product/api) - <https://www.semanticscholar.org/product/api>
- [Crossref REST API documentation](https://www.crossref.org/documentation/retrieve-metadata/rest-api/) - <https://www.crossref.org/documentation/retrieve-metadata/rest-api/>
- [Example Google Scholar MCP Server repository](https://github.com/JackKuo666/Google-Scholar-MCP-Server) - <https://github.com/JackKuo666/Google-Scholar-MCP-Server>
- [Example Google Scholar Search MCP listing](https://mcpservers.org/servers/lwaetzig/google-scholar-search-mcp) - <https://mcpservers.org/servers/lwaetzig/google-scholar-search-mcp>
- [gs-skills listing](https://skilld.dev/skills/cookjohn/gs-skills/gs-navigate-pages) - <https://skilld.dev/skills/cookjohn/gs-skills/gs-navigate-pages>
- [Survey on Class-Agnostic Counting](https://arxiv.org/abs/2501.19184) - <https://arxiv.org/abs/2501.19184>
- [Zero-Shot Object Counting, CVPR 2023](https://openaccess.thecvf.com/content/CVPR2023/papers/Xu_Zero-Shot_Object_Counting_CVPR_2023_paper.pdf) - https://openaccess.thecvf.com/content/CVPR2023/papers/Xu_Zero-Shot_Object_Counting_CVPR_2023_paper.pdf
- [TAO Dataset](https://taodataset.org/) - <https://taodataset.org/>
- [Example 3D counting paper](https://arxiv.org/html/2411.19149v2) - <https://arxiv.org/html/2411.19149v2>

10. Final takeaway

You do not need to manually inspect 10,000 Google Scholar results. Use a modality-first Google Scholar strategy to collect 55-85 strong seed papers, then use Claude Code or Codex with legitimate scholarly APIs to expand, deduplicate, organize, and prepare the literature matrix. Google Scholar remains useful for seed discovery, “Cited by,” “Related articles,” and verification, but not for bulk automated retrieval.